

a Locked prerequisites

T002 one certificate per distinct prime;
repeats collapse

T003 S-unit status survives extension,
product, and inverse

T004 rational log sums close to $\log q$;
 q^2 remains an S-unit

b Relation-to-capacity proof spine

T005

PROVED

rational relation in $\{v_p\}$
 $\Rightarrow$ integer coefficients m_p

T006

PROVED

substitute the hits:
 $\log R = \beta$, with $R > 0$ and
 R, β algebraic

T007

PROVED

Hermite--Lindemann:
 $\beta = 0$ and $R = 1$

T010

PROVED

$\{v_p : p \notin S\}$ is independent
 $\Rightarrow |P_{\text{hit}}| \leq \dim V + |S|$

T009

PROVED

at $w \mid p$, $p \notin S$:
 $2m_p v_w(p) = 0 \Rightarrow m_p = 0$

T008

PROVED

square in one number field:
 $\prod p^{2m_p} = \prod (q_p^2)^{m_p}$

Machine ledger

proof IDs 20

dependency cycles 0

controls 6

target matches 0

Highlighted valuation isolation is the arithmetic step that eliminates every nonzero outside-prime coefficient.